

SECOND PUBLIC EXAMINATION

Honour School of Mathematics Part C: Paper SC7
Honour School of Mathematics and Statistics Part C: Paper SC7
Honour School of Mathematics and Computer Science Part C: Paper SC7
Master of Science in Mathematical Sciences: Paper SC7

BAYES METHODS

TRINITY TERM 2021

Thursday 3 June

Solutions

Do not turn this page until you are told that you may do so

1. For $n \geq 1$ let $V_n = \{1, \dots, n\}$ and let $\mathcal{E}_n = \{(i, j) : 1 \leq i < j \leq n\}$ be the set of all undirected edges (i, j) . For $E \subseteq \mathcal{E}_n$ let (V, E) be a graph with vertex-set V and edge-set E .

For $i = 1, \dots, n$ let $x_i \in [0, 1]^2$ be n distinct points in the unit square and let $x = (x_1, \dots, x_n)$. For $(i, j) \in E$ let $\ell_{(i,j)}$ be the edge-segment connecting x_i and x_j , so that

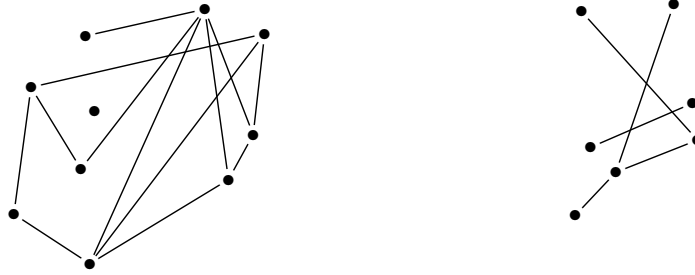
$$\ell_{(i,j)} = \{x_i + c(x_j - x_i) : 0 < c < 1\}.$$

For given $x, (V, E)$, let

$$L(x, V, E) = \{\ell_{(i,j)}\}_{(i,j) \in E}$$

be the set of line segments determined in this way. We call $Y = L(x, V, E)$ a “straight-line-graph”. Let $\mathcal{G}_x = \cup_{E \subseteq \mathcal{E}_n} \{L(x, V, E)\}$ be the set of all straight-line graphs on x . Let $\mathbb{I}\{A\}$ be the indicator for A and let $\#Y$ equal the number of “edge-segment crossings”,

$$\#Y = \frac{1}{2} \sum_{\ell \in Y} \sum_{\substack{\ell' \in Y \\ \ell' \neq \ell}} \mathbb{I}\{\ell \cap \ell' \neq \emptyset\}.$$



Above, two straight line graphs: (left) $n = 10$ and $\#Y = 6$; (right) $n = 7$ and $\#Y = 3$.

- (a) [9 marks] Suppose $x = (x_1, \dots, x_n)$ is fixed. Consider the observation model

$$p(y|\theta, x) = \exp(-\theta \#y) / Z(\theta, x), \quad y \in \mathcal{G}_x,$$

for $\theta > 0$ a parameter penalising crossings and $Z(\theta, x)$ a normalising constant. In the following we drop “ x ” dependence from the notation when it is fixed.

Let $Y \sim p(\cdot|\theta)$, $Y \in \mathcal{G}_x$ and suppose we observe $Y = y_{obs}$. A prior $\pi(\theta)$ for θ is given.

- (i) Write down the posterior for $\theta|y_{obs}$ and explain why it is doubly intractable.

Solution: (2/B/S)

$$\pi(\theta|y_{obs}) = p(y_{obs}|\theta)\pi(\theta)/p(y_{obs})$$

with $p(y_{obs}) = \int_0^\infty p(y_{obs}|\theta)\pi(\theta)d\theta$. This is DI as

$$\frac{p(y_{obs}|\theta)}{p(y_{obs}|\theta')} = \frac{Z(\theta')}{Z(\theta)} \exp((\theta - \theta')\#y_{obs}),$$

and we cant evaluate this ratio as $Z(\theta)$ is intractable.

Remark (following not essential): It is given by

$$Z(\theta) = \sum_{y \in \mathcal{G}_x} \exp(-\theta \#y),$$

and we are told this is intractable (it is, for n at all large).

- (ii) Consider Approximate Bayesian Computation (ABC). Specify and briefly justify a summary statistic $S(y)$, $y \in \mathcal{G}_x$ and write down the ABC posterior $\pi_{ABC}(\theta|y_{obs})$ in terms of the model elements $\pi(\theta)$, $p(y|\theta)$ and $S(y)$ and a threshold $\delta > 0$.

Solution: (4/S) Take $S(y) = \#y$, $y \in \mathcal{G}_x$. This is actually sufficient so we cannot do better. Let $\Delta_\delta(y_{obs}) = \{y \in \mathcal{G}_x : |S(y) - S(y_{obs})| < \delta\}$.

$$\pi_{ABC}(\theta|y_{obs}) = \frac{p(\Delta_\delta(y_{obs})|\theta)\pi(\theta)}{p(\Delta_\delta(y_{obs}))},$$

with

$$p(\Delta_\delta(y_{obs})|\theta) = \sum_{y \in \Delta_\delta(y_{obs})} p(y|\theta)$$

and

$$p(\Delta_\delta(y_{obs})) = \int_0^\infty p(\Delta_\delta(y_{obs})|\theta)\pi(\theta)d\theta.$$

- (iii) Suppose an algorithm simulating $Y \sim p(\cdot|\theta)$ is available. Give an ABC-rejection algorithm targeting $\pi_{ABC}(\theta|y_{obs})$.

Solution: (3/B/S) Let $\delta > 0$ be given.

1. simulate $\theta \sim \pi(\cdot)$ and $y \sim p(\cdot|\theta)$.
2. if $|S(y_{obs}) - S(y)| < \delta$ stop and return θ and otherwise goto 1.

- (b) [9 marks] Give an MCMC algorithm targeting $Y \sim p(\cdot|\theta)$ and show it is irreducible. [Hint: choose an edge at random from $\mathcal{E}_n$; add it if absent and otherwise remove it.]

Solution: (9/S/N) The only thing that actually varies between $y, y' \in \mathcal{G}_x$ is E . Suppose $X_t = E$ for some $E \subseteq \mathcal{E}_n$.

1. Select (i, j) uniformly at random from $\mathcal{E}_n$, If $(i, j) \in E$ set $E' = E \setminus \{(i, j)\}$ and otherwise set $E' = E \cup \{(i, j)\}$.
2. Let $G = L(x, V, E)$ and $G' = L(x, V, E')$.
3. with probability

$$\alpha(E'|E) = \min \{1, \exp(\theta(\#G - \#G'))\}$$

set $X_{t+1} = E'$ and otherwise set $X_{t+1} = E$.

(ergodic) This is irreducible as $\alpha(E'|E) > 0$ for E, E' differing by one edge. Starting from E the probability to choose and delete in sequence the edges in $E \setminus E'$ and then choose and add in sequence the edges in $E' \setminus E$ is greater than zero, so there is a path from E to E' .

Remarks not required: It is aperiodic as it is irreducible and we can reject. It is stationary wrt $p(\cdot|\theta)$ as it is a MH algorithm, and so it is ergodic as $\mathcal{G}_x$ is finite. The Markov chain $G_t = L(x, V, X_t)$, $t = 0, 1, 2, \dots$ of straight line graphs might be initialised with $X_0 = \emptyset$. The proposal probability $q(E'|E) = \binom{n}{2}^{-1}$ cancels in the

Hastings ratio as $q(E'|E) = q(E|E')$.

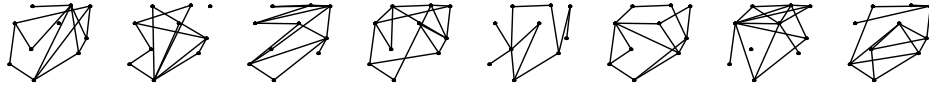


Figure above illustrative and not required.

- (c) [7 marks] Let $\theta > 0$ and $q \in (0, 1)$ be given. For $n \in \{0, 1, 2, \dots\}$, $x \in \mathbb{R}^{2n}$, $y \in \mathcal{G}_x$, and $f(x)$ a given function of x , consider the probability distribution

$$p(n, x, y|q, \theta) \propto f(x) \exp(-\theta \#y) n! q(1-q)^{n-1}.$$

Here $f(x)$ is a positive function ensuring $p(n, x, y|q, \theta)$ is finitely normalisable.

Give the updates adding and deleting points from x in reversible jump MCMC targeting $p(n, x, y|q, \theta)$. [Hint: one point in the figure above would be particularly easy to delete.]

Solution: (7/N) Suppose $X_t = (n, x, y)$ with $y \in \mathcal{G}_x$. Given x and y the values of E and V can be recovered. Following the hint we only add and delete points if they have no edges connected to them. We can only delete points we could add. We reject updates that propose a deletion at an edge-connected point (for simplicity).

1. with probability $1/2$ choose to add a point and otherwise delete.
2. (if we choose to add) simulate $x'_{n+1} \sim U(0, 1)^2$. Set $n' = n + 1$ and $x' = x \cup x_{n+1}$. The set of line segments does not change if we add a point with no edges, so set $y' = y$. With probability

$$\begin{aligned} \alpha(n', x', y'|n, x, y) &= \min \left\{ 1, \frac{f(x') \exp(-\theta \#y') (n+1)! q(1-q)^{n'-1} \times 0.5 \frac{1}{n+1}}{f(x) \exp(-\theta \#y) n! q(1-q)^{n-1} \times 0.5} \right\} \\ &= \min \left\{ 1, \frac{f(x')}{f(x)} (1-q) \right\}, \end{aligned}$$

set $X_{t+1} = (n', x', y')$ and otherwise set $X_{t+1} = (n, x, y)$.

3. (if we choose to delete) choose a point $i \sim U\{1, \dots, n\}$. If $n = 1$ or there are any edges connected to vertex i in V then set $X_{t+1} = (n, x, y)$. Otherwise set $n' = n - 1$, $x' = x \setminus \{x_i\}$ and $y' = y$. With probability

$$\alpha(n', x', y'|n, x, y) = \min \left\{ 1, \frac{f(x')}{f(x)} (1-q)^{-1} \right\}$$

set $X_{t+1} = (n', x', y')$ and otherwise set $X_{t+1} = (n, x, y)$.

Remark (not needed): we choose $f(x)$ to control the explosion of terms; students should not need to consider this at all; there would be additional updates adding and deleting edges.

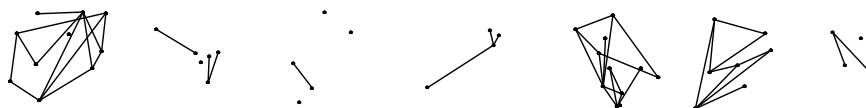


Figure above illustrative and not required.

2. When farming began, it is thought smaller settlements farmed more intensively, and in particular fertilized their crops more heavily. Archaeologists interested in testing this hypothesis characterised fertilizer levels as "low" (0), "medium" (1) or "high" (2). Fertilizer level changes the nitrogen isotope ratio in seeds from plants. The archeologists gathered n seeds from excavations at settlement sites with different settlement area sizes.

For $i = 1, \dots, n$ let $x_i > 0$ give the area of the settlement where the i 'th seed was excavated and let $y_i \in \mathbb{R}$ give the nitrogen isotope ratio measurement. This is normal with a mean $\mu_{f_i} = E(y_i)$, $f_i \in \{0, 1, 2\}$ and $-\infty < \mu_0 < \mu_1 < \mu_2 < \infty$ and standard deviation $\sigma > 0$.

A model for the unknown fertiliser level $f_i \in \{0, 1, 2\}$ the plants received is given in terms of a latent variable $z_i \in \mathbb{R}$, $i = 1, \dots, n$ and two unknown thresholds, $-\infty < c_1 < c_2 < \infty$. The observation model is as follows

$$\begin{aligned} z_i &\sim N(\beta x_i, 1) \\ f_i &= \mathbb{I}_{z_i > c_1} + \mathbb{I}_{z_i > c_2} \\ y_i &\sim N(\mu_{f_i}, \sigma^2), \text{ all jointly independent for } i = 1, \dots, n. \end{aligned}$$

The archaeologists have some modern isotope ratio data v_j , $j = 1, \dots, m$, in which the manure levels $g_j \in \{0, 1, 2\}$, $j = 1, \dots, m$ are known but there are no settlement-area values. The observation model for the modern data is

$$v_j \sim N(\mu_{g_j}, \sigma^2), \text{ jointly independent for } j = 1, \dots, m.$$

- (a) [10 marks] Let $y = (y_1, \dots, y_n)$, $z = (z_1, \dots, z_n)$, $v = (v_1, \dots, v_m)$ and $g = (g_1, \dots, g_m)$. Let $c = (c_1, c_2)$ and $\mu = (\mu_0, \mu_1, \mu_2)$.
- (i) What factors would you consider in choosing a prior for β and how would you check it represented available prior knowledge?

Solution: (4/S) The main point (1) would be to give it a density symmetric about zero (so non-informative with respect to the key scientific hypothesis). The scale (2) would depend on the scale of x . Prior preference for the proportions (3) of low/medium/high manuring levels would play a role, but this would interact with the c -prior. Simulation (4) might be used to check these were acceptable. This would lead to a test for $\beta < 0$ against $\beta \geq 0$. Other considerations would be acceptable.

- (ii) Suppose a joint prior $\pi(\beta, c, \mu, \sigma)$ is given. Write down the posterior $\pi(\beta, c, \mu, \sigma, z|y, v, g)$ in terms of the model elements, up to a constant factor.

Solution: (4/S) We have

$$\pi(\beta, c, \mu, \sigma, z|y, v, g) \propto \pi(\beta, c, \mu, \sigma) p(v|\mu, \sigma, g) p(y|\mu, \sigma, z, c) p(z|\beta)$$

where

$$p(y|\mu, \sigma, z) = \prod_{i=1}^n N(y_i; \mu_{f_i}, \sigma^2),$$

where $f_i = f_i(z_i, c)$ as in the question and

$$p(v|\mu, \sigma, g) = \prod_{j=1}^m N(v_j; \mu_{g_j}, \sigma^2),$$

and

$$p(z|\beta) = \prod_{i=1}^n N(z_i; \beta x_i, 1).$$

- (iii) How would you test the hypothesis raised by the archaeologists?

Solution: (2/S/N) Calculate the posterior odds for $\beta < 0$. This would equal the Bayes factor if we took the β prior symmetric and if not we divide by the prior odds for $\beta < 0$. The point here is that if $\beta < 0$ then z tends to take lower values for larger sites and *v.v.* and so we would see higher manuring levels at smaller sites.

- (b) [15 marks] Suppose we take a simple approach, estimating $\hat{\mu}_0, \hat{\mu}_1, \hat{\mu}_2$ and $\hat{\sigma}$ with their MLE's from the modern data v, g and then setting

$$\hat{f}_i = \arg \max_{k=0,1,2} N(y_i; \hat{\mu}_k, \hat{\sigma}^2)$$

for $i = 1, \dots, n$. Let $\hat{f} = (\hat{f}_1, \dots, \hat{f}_n)$. These $\hat{f}$ -estimates are now taken as fixed data.

- (i) Consider estimating β with an estimate of the posterior mean $E_{z,c,\beta|\hat{f}}(\beta)$. Identify a weakness of this approach. Give a setting where it would nevertheless lead to good approximate inference for β .

Solution: (4/N) If our loss for estimation of β is Mean Square Error (as it presumably is given the posterior mean) and we want to minimise the Bayes Risk then we would take the posterior mean in the distribution we wrote down in previous part, *ie* $\mu_\beta = E_{z,c,\beta,\mu,\sigma|y,v,g}(\beta)$. The approach suggested will not minimise the Bayes Risk. It doesn't fully quantify uncertainty. If the modern data set was very large so estimates $\hat{\mu}, \hat{\sigma}$ were very accurate, and the distributions $N(\hat{\mu}_k, \hat{\sigma}^2)$ had little overlap for $k = 0, 1, 2$ then the values of f_i in the full posterior above would vary very little away from $\hat{f}_i$, so the marginal posterior for β would be the same or near in both cases.

- (ii) Write down the posterior $\pi(z, c, \beta|\hat{f})$ in terms of the model elements and a prior $\pi(c, \beta)$ for c and β , up to a constant factor.

Solution: (2/S/N)

$$\pi(z, c, \beta|\hat{f}) \propto p(z|c, \beta, \hat{f})\pi(c, \beta),$$

with

$$p(z|c, \beta, \hat{f}) \propto \prod_{i=1}^n N(z_i; \beta x_i, 1) \mathbb{I}(\hat{f}_i = \mathbb{I}_{z_i > c_1} + \mathbb{I}_{z_i > c_2}).$$

- (iii) Calculate the marginal posterior $\pi(\beta, c|\hat{f})$ giving your answer as an explicit function of c, β and $\hat{f}$ and $x_1, \dots, x_n$, and involving the CDF Φ of the standard normal.

Solution: (6/N) The integrals over z_i , $i = 1, \dots, n$ factorise at fixed c, β so we

consider

$$\begin{aligned}\pi(\beta, c|\hat{f}) &= \int_{\mathbb{R}^n} \pi(z, c, \beta|\hat{f}) dz \\ &\propto \pi(c, \beta) \prod_{i=1}^n \int_{-\infty}^{\infty} \tilde{p}(z_i|c, \beta, \hat{f}_i) dz_i\end{aligned}$$

with

$$\tilde{p}(z_i|c, \beta, \hat{f}_i) = N(z_i; \beta x_i, 1) \mathbb{I}(\hat{f}_i = \mathbb{I}_{z_i > c_1} + \mathbb{I}_{z_i > c_2}).$$

Now

$$\begin{aligned}\int_{-\infty}^{\infty} \tilde{p}(z_i|c, \beta, \hat{f}_i) dz_i &= \int_{-\infty}^{\infty} N(z_i; \beta x_i, 1) \left[\mathbb{I}(\hat{f}_i = \mathbb{I}_{z_i > c_1} + \mathbb{I}_{z_i > c_2}) \right] dz_i \\ &= \int_{-\infty}^{c_1} N(z_i; \beta x_i, 1) \mathbb{I}(\hat{f}_i = 0) dz_i + \int_{c_1}^{c_2} N(z_i; \beta x_i, 1) \mathbb{I}(\hat{f}_i = 1) \\ &\quad + \int_{c_2}^{\infty} N(z_i; \beta x_i, 1) \mathbb{I}(\hat{f}_i = 2) dz_i\end{aligned}$$

and since $\int_{-\infty}^{c_1} N(z_i; \beta x_i, 1) dz_i = \int_{-\infty}^{c_1 - \beta x_i} N(u_i; 0, 1) du_i$

$$\begin{aligned}&= \Phi(c_1 - \beta x_i) \mathbb{I}(\hat{f}_i = 0) + [\Phi(c_2 - \beta x_i) - \Phi(c_1 - \beta x_i)] \mathbb{I}(\hat{f}_i = 1) \\ &\quad + \Phi(\beta x_i - c_2) \mathbb{I}(\hat{f}_i = 2)\end{aligned}$$

- (iv) Given a choice between carrying out the inference using MCMC targeting $\pi(z, c, \beta|\hat{f})$ or MCMC targeting $\pi(c, \beta|\hat{f})$, what factors would you consider in making a choice?

Solution: (3/S/N) We would generally work with the marginals as the MCMC will typically converge more rapidly. If the function evaluations involved in the marginal were intractable or time-consuming we might prefer the joint, however that is not the case here. There is a convenient Gibbs sampler for z which may make the joint competitive. We would take the joint if the z 's were needed in the inference. However the inference is based on β and the z 's are auxiliary variables which play no role beyond defining the model. So on balance the marginal but perhaps some case for the joint.

3. (a) [11 marks] For $p \geq 1$, real $\alpha > 0$ and H a continuous distribution over $\Omega = \mathbb{R}^p$, a random distribution $G \sim \Pi(\alpha, H)$ on Ω has a Dirichlet Process distribution with parameter α and base distribution H . Given G and $n \geq 1$, parameters $\theta_i \sim G$, $i = 1, 2, \dots, n$ are jointly independent and distributed according to G . Let $\theta = (\theta_1, \dots, \theta_n)$.

- (i) The Multinomial Dirichlet Process and the Dirichlet process are related. State the relation and what insight it gives us as to the character of the Dirichlet process.

Solution: (2/B) The multinomial DP converges to the DP. Let G_M be a realisation of the MDP. We wrote that for any suitable (ie measurable) partition $A_1, \dots, A_r$ of Ω the joint density of $G_M(A_1), \dots, G_M(A_r)$ converges to the joint density of $G(A_1), \dots, G(A_r)$. This gives us the insight that the DP is atomic, with an infinite number of weighted atoms distributed over Ω .

- (ii) State briefly why the Dirichlet process may be useful for clustering.

Solution: (3/B/S) Precisely because it is atomic, we can take one parameter θ_i for each observation y_i , $i = 1, \dots, n$. The DP allows $\theta_i = \theta_j$ with a probability that is greater than zero. In this way observations with equal parameters are grouped into clusters.

- (iii) Suppose $n \geq 10$. Explain how to simulate θ_9 and θ_{10} without simulation of $G \sim \Pi(\alpha, H)$.

Solution: (3/S) By exchangeability of $\theta_1, \dots, \theta_n, \dots$, we can take these first in the process. In the marginal process we simulate $\theta_9 \sim H$ and then with probability $1/(\alpha + 1)$ we set $\theta_{10} = \theta_9$ and otherwise we simulate $\theta_{10} \sim H$.

- (iv) Calculate the probability that $\theta_7 = \theta_8$ and (in the same realisation) $\theta_9 = \theta_{10}$.

Solution: (3/S) By exchangeability this is the same as the probability that $\theta_1 = \theta_2$ and (in the same realisation) $\theta_3 = \theta_4$. This can happen two ways. Either 1 and 2 are in the same cluster S_1 and 3 and 4 are in S_2 or they can all be in S_1 . From the CRP, in a format that reflects the arrival orders,

$$\Pr(\theta_1 = \theta_2, \theta_3 = \theta_4) = \frac{1}{(\alpha + 1)} \frac{\alpha}{(\alpha + 2)} \frac{1}{(\alpha + 3)} + \frac{1}{(\alpha + 1)} \frac{2}{(\alpha + 2)} \frac{3}{(\alpha + 3)}.$$

- (b) [14 marks] Suppose now we have two independent Dirichlet processes $G_1 \sim \Pi(\alpha_1, H_1)$ and $G_2 \sim \Pi(\alpha_2, H_2)$ with $\alpha_1, \alpha_2 > 0$ and H_1 and H_2 continuous distributions over $\mathbb{R}$ with densities h_1 and h_2 respectively.

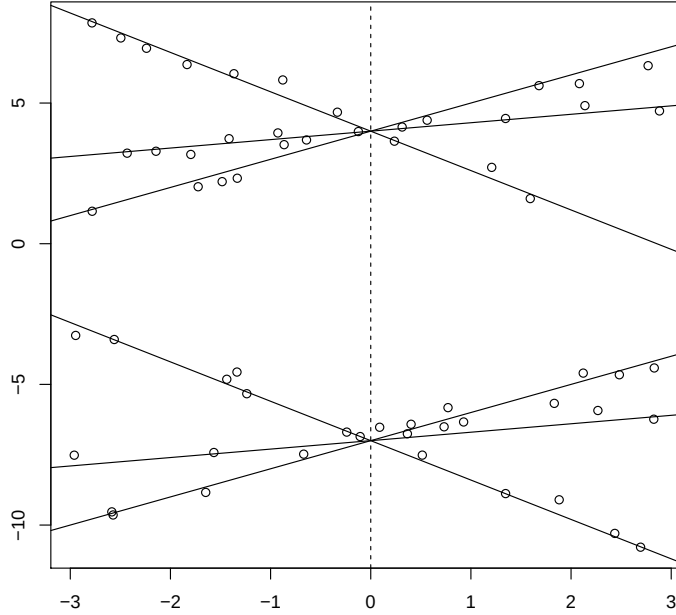
Let $\theta_{1,j} \sim G_1$ and $\theta_{2,j} \sim G_2$ all jointly independent given G_1 and G_2 for $j = 1, \dots, n$. For $i = 1, 2$, let $\theta_i = (\theta_{i,1}, \dots, \theta_{i,n})$. For $j = 1, \dots, n$ let $x_j \in \mathbb{R}$ be a scalar covariate. Consider the observation model

$$y_j = \theta_{1,j} + \theta_{2,j}x_j + \epsilon_j, \quad \epsilon_j \sim N(0, \sigma^2),$$

with ϵ_j jointly independent for $j = 1, \dots, n$. A prior $\pi(\sigma)$ for σ is given.

- (i) Explain qualitatively, perhaps with the aid of a graph plotting y against x , what sort of clustering is modelled here.

Solution: (5/N) The data are partitioned into groups by intercept and slope. If there are K_1 unique values among $\theta_{1,j}$, $j = 1, \dots, n$ and K_2 unique values among $\theta_{2,j}$, $j = 1, \dots, n$ then we have $K_1 \times K_2$ regression lines. The data are partitioned over these lines. There are K_1 different intercepts. For each intercept there are K_2 different slopes. The set of slopes paired with each intercept is the same for each intercept. Here is a figure illustrating this.



- (ii) For $i = 1, 2$ let $\theta_i^* = (\theta_{i,1}^*, \dots, \theta_{i,K_i}^*)$ give the K_i unique parameter values in $\theta_i = (\theta_{i,1}, \dots, \theta_{i,n})$. For $i = 1, 2$ let $S_i = (S_{i,1}, \dots, S_{i,K_i})$ where for $k = 1, \dots, K_i$,

$$S_{i,k} = \{j : \theta_{i,j} = \theta_{i,k}^*, j = 1, \dots, n\}.$$

Give the posterior distribution $\pi(\theta_1^*, S_1, \theta_2^*, S_2, \sigma | y)$ in terms of the model elements up to a constant factor. There is no need to write the prior distributions of S_1 and S_2 explicitly, but you should identify them with a known distribution.

Solution: (3/S/N) For $i = 1, 2$ let

$$\pi_i(\theta_i^* | S_i) = \prod_{k=1}^{K_i} h_i(\theta_{i,k}^*)$$

$$p(y | \theta_1^*, S_1, \theta_2^*, S_2, \sigma) = \prod_{j=1}^n N(y_j; \theta_{1,k_{1,j}}^* + \theta_{2,k_{2,j}}^* x_j, \sigma^2)$$

where for $j = 1, \dots, n$ and $i = 1, 2$ we set $k_{i,j} = \{k : j \in S_{i,k}, j = 1, \dots, n\}$. The posterior is

$$\pi(\theta_1^*, S_1, \theta_2^*, S_2, \sigma | y) \propto p(y | \theta_1^*, S_1, \theta_2^*, S_2, \sigma) \pi(\sigma) \prod_{i=1,2} \pi_i(\theta_i^* | S_i) P_{\alpha_i, n}(S_i).$$

Here for $i = 1, 2$, we have $P_{\alpha_i, n}(S_i)$ equal to the CRP distribution over partitions S_i of the set $\{1, \dots, n\}$.

- (iii) Give the cluster-membership update in a Gibbs sampler targeting $\pi(\theta_1^*, S_1, \theta_2^*, S_2, \sigma | y)$. It is sufficient to give the update for the cluster assignment of j in S_1 for one $j \in \{1, \dots, n\}$.

Solution: (6/S/N) We have two CRP's which run independently, so this is essentially unchanged from notes, just with an extra subscript. Let S_1^{-j} be the partition with j removed. Let $\theta_{-j,1}^*$ be the corresponding parameter vector. Let K_1^{-j} be the number of clusters in S_1^{-j} . Let $\theta_{-j,1,K_1^{-j}+1}^* \sim h_1$ be an independent draw from h_1 . The distribution we require is, for $k = 1, \dots, K_1^{-j}, K_1^{-j} + 1$,

$$\begin{aligned} & \Pr(j \in S_{1,k}^{-j} | S_1^{-j}, \theta_{-j,1}^*, \theta_{-j,1,K_1^{-j}+1}^*, \theta_2^*, S_2, y) \\ & \propto \begin{cases} n_{1,k}^{-j} N(y_j; \theta_{-j,1,k}^* + \theta_{2,k_{2,j}}^* x_j, \sigma^2) & \text{for } k = 1, \dots, K_1^{-j} \text{ and} \\ \alpha N(y_j; \theta_{-j,1,k}^* + \theta_{2,k_{2,j}}^* x_j, \sigma^2) & \text{when } k = K_1^{-j} + 1, \end{cases} \end{aligned}$$

where $k_{2,j} = \{k : j \in S_{2,k}, j = 1, \dots, n\}$ and $n_{1,k}^{-j} = |S_{1,k}^{-j}|$ for $k = 1, \dots, K_1^{-j}$ is the number of elements in $S_{1,k}^{-j}$.

In order to carry out the update, remove j from S_1 to form S_1^{-j} , calculate $\Pr(j \in S_{1,k}^{-j} | S_1^{-j}, \theta_{-j,1}^*, \theta_{-j,1,K_1^{-j}+1}^*, \theta_2^*, S_2, y)$ for $k = 1, \dots, K_1^{-j}, K_1^{-j} + 1$ by normalising the expressions above, and sample k accordingly. Then assign j to cluster k in S_1^{-j} , creating a new cluster if $k = K_1^{-j} + 1$. If $k = K_1^{-j} + 1$ set $\theta_1^* = (\theta_{-j,1,1}^*, \dots, \theta_{-j,1,K_1^{-j}+1}^*)$ and otherwise $\theta_1^* = \theta_{-j,1}^*$.